Energy skate park simulation

National 5 · Engineering contexts & challenges · Answers

Answers and marking guidance

Working is shown for each answer. Other valid methods should be credited.

1 Place the skater at the top of the ramp and release them. What happens?

- The skater travels down the ramp and up the other side, speeding up then slowing down.

2 Turn on the bar graphs and pie chart. What do they show?

- The total energy and how much of it is potential, kinetic and (with friction) thermal energy.

3 What do you notice about the total energy at each point on the track?

- It stays the same (with no friction).

4 How does the skater's speed relate to their kinetic energy?

- The faster the skater moves, the more kinetic energy they have.

5a) their potential energy?

- It increases ($E_p = mgh$).

5b) their kinetic energy?

- It increases ($E_k = ½mv^2$).

6 Add friction to the track. What happens to the energy over time?

- Potential and kinetic energy fall as energy is changed into heat by friction, until the skater stops.
- The total energy, including the heat, stays the same.

7 In the Playground tab, design a track where the skater reaches maximum speed without flying off. Sketch or describe it.

- A high, steep starting drop gives the most speed.
- A smooth curve at the bottom, with the rising section no higher than the start, keeps the skater on the track.

8 What design features helped reduce energy losses?

- No friction, smooth curves and no sharp corners.

9 Explain, in your own words, how energy is conserved in the simulation.

- Energy can't be created or destroyed, only changed from one form to another: potential becomes kinetic and back again, and the total stays constant.

10 Why is conservation of energy important for engineers?

- It lets engineers track where energy goes, reduce losses and design more efficient systems, saving cost and resources.